Funded by
DFG Deutsche Forschungsgemeinschaft
German Research Foundation

Universität Potsdam

Combining Satellite and In-Situ Data for Ecosystem Monitoring

February 23, 2026

Today's Schedule

Day 2: Contextualizing in-situ data with satellite remote sensing

09:30 – 10:00 Introduction to Google Earth Engine for data processing

10:00 – 10:30 Practical Exercise – Visualizing data using online tools

10:30 – 10:45 Tea/Coffee Break

10:45 – 11:15 Practical Exercise – Monitoring changes in high-elevation regions

11:15 – 12:30 Practical Example – Extracting Rainfall time series for specific locations

12:30 – 13:30 Lunch Break

13:30 – 14:30 Practical Example – Integrating watershed data and mapping climate trends

14:30 – 14:45 Tea/Coffee Break

14:45 – 15:45 Practical Example – Mapping changes in high-altitude rivers

15:45 – 16:00 Research Outlook, Ongoing Work, and Wrap Up

Shared Text

I will include a couple of web links in this presentation – to access them more smoothly, I will copy/paste the text here:

https://sharepad.io/p/9KdFEro

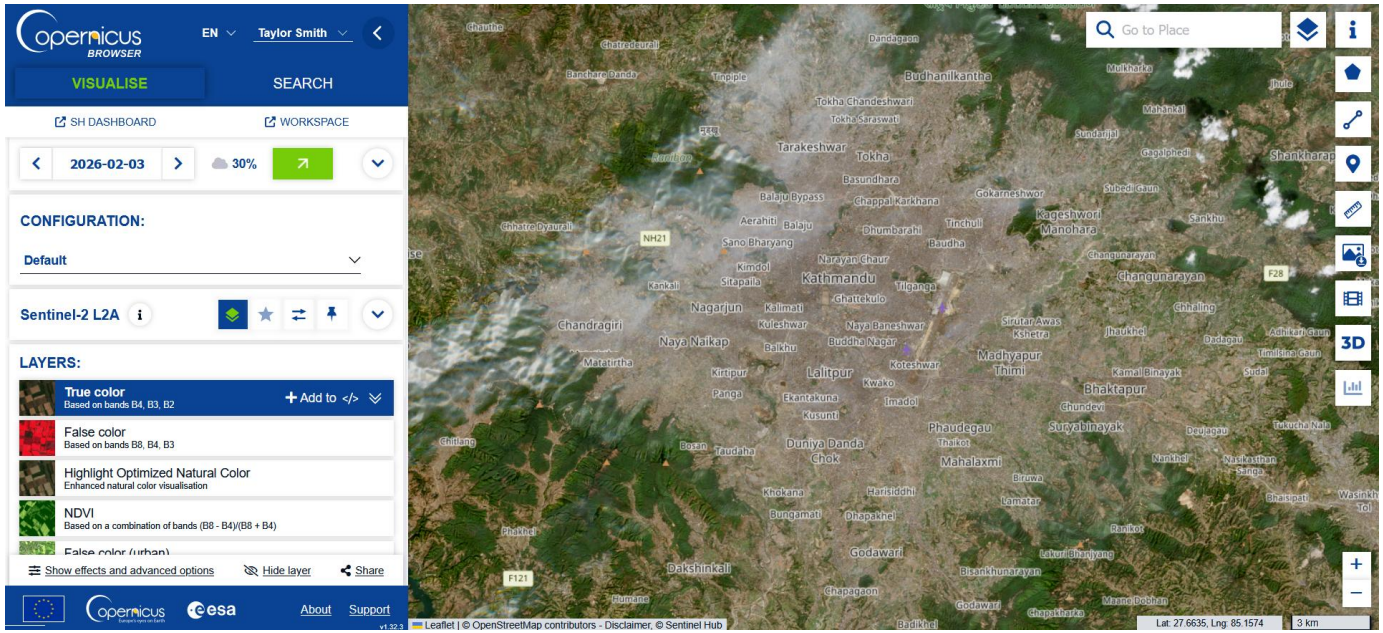
Why Online Processing?

- No Need for Large Computer
- Works with low internet access
- Many data sets available
- Can be accessed using programming languages or via a web browser

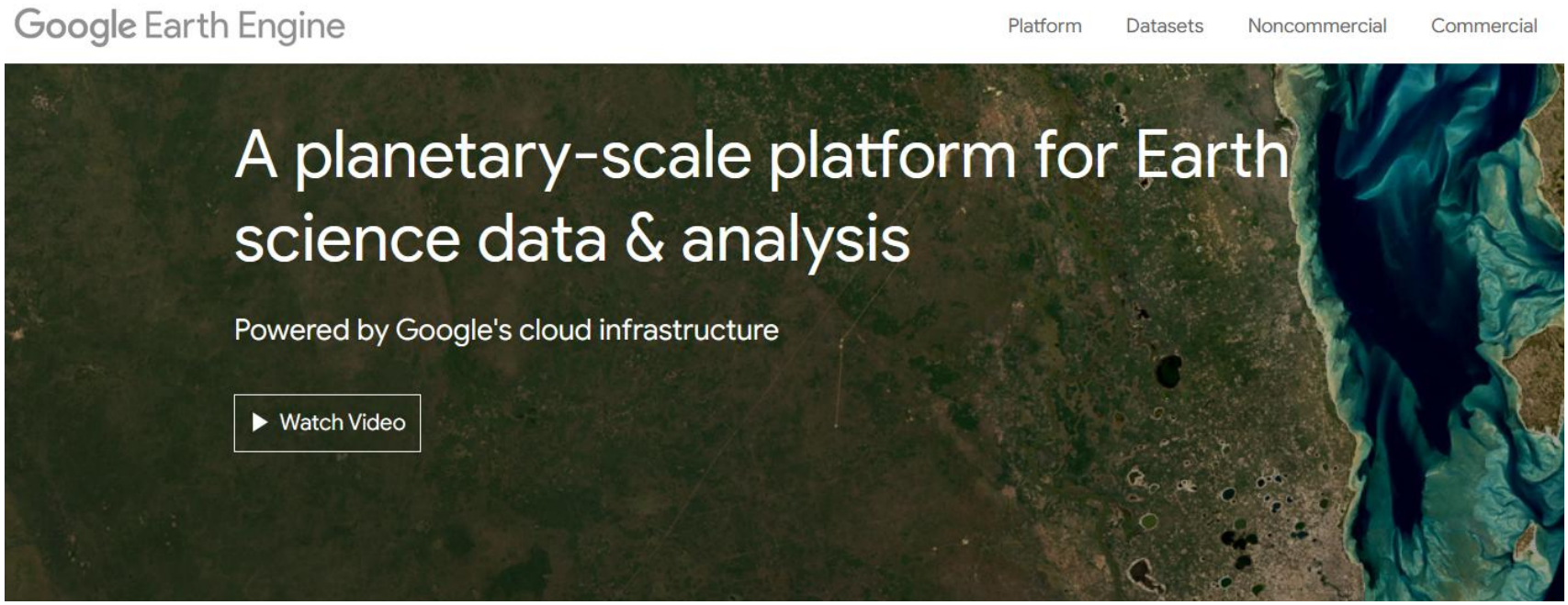
Introduction to Online Data Processing

Quick Demo: Copernicus Browser

https://browser.dataspace.copernicus.eu/?zoom=12&lat=27.69526&lng=85.34145&themeId=DEFAULT-THEME&visualizationUrl=U2FsdGVkX1%2BaTHT9zOW5sXEwkg1D2Rt5LvJZkHfL%2FuL94du0%2BOp4MuOhiH%2BnmyduBivGeWLzW8RoejzyaktHk1u3ABBfovuK0OfpziFGwoj93Vppb0VzQby9b0gTngxo&datasetId=S2_L2A_CDAS&fromTime=2026-02-03T00%3A00%3A00.000Z&toTime=2026-02-03T23%3A59%3A59.999Z&layerId=1_TRUE_COLOR&demSource3D=%22MAPZEN%22&cloudCoverage=30&dateMode=SINGLE



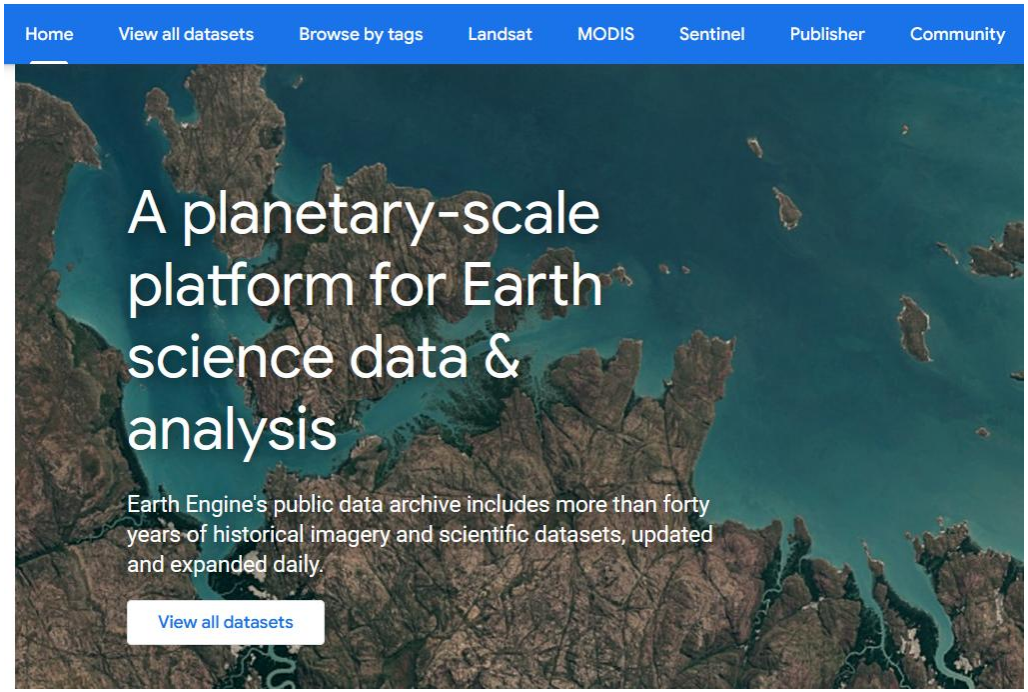
Google Earth Engine



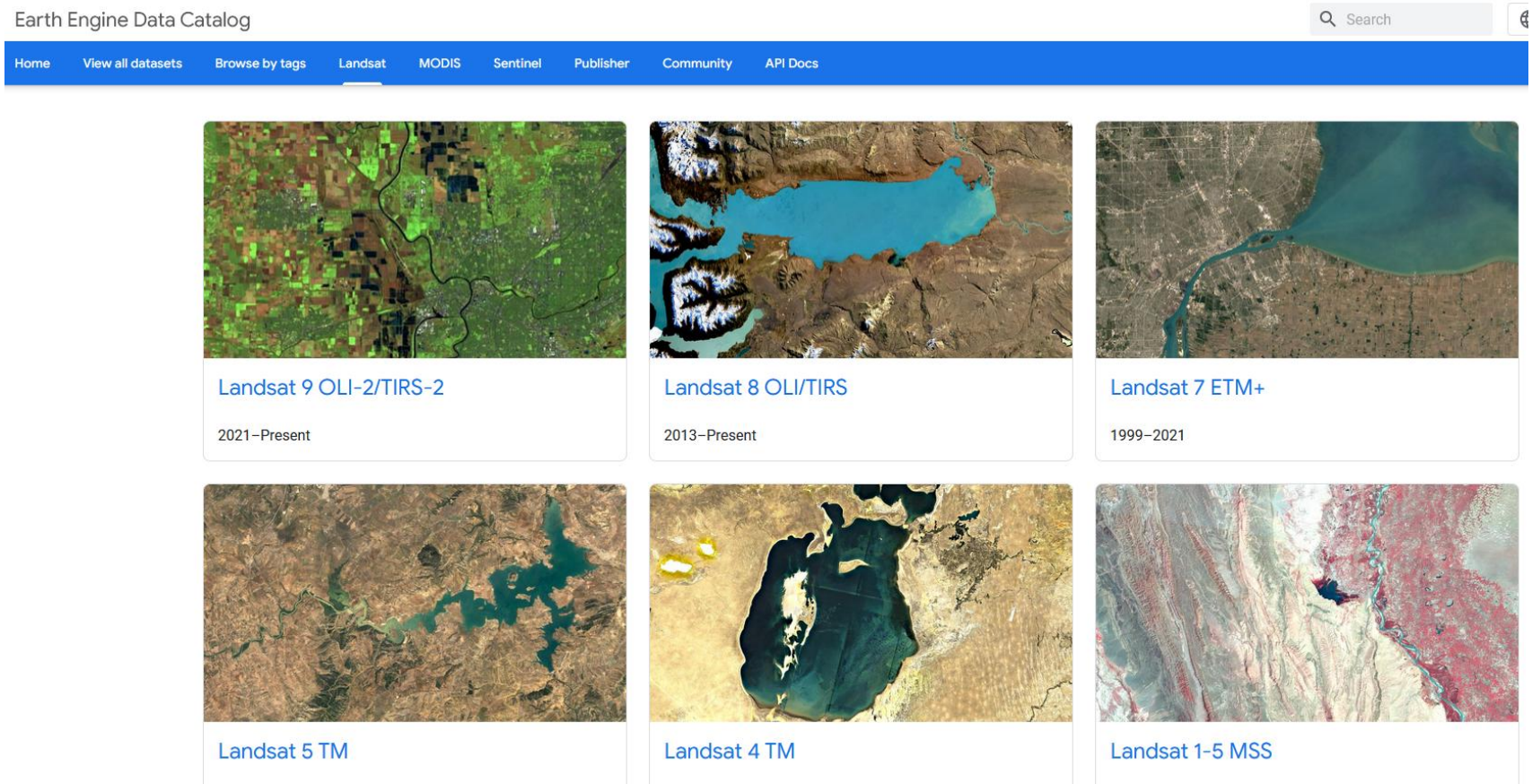
Google Earth Engine

- Launched in 2010 at COP16
- Managed by Google
- Free for Non-Commercial use
- Also available for commercial customers
- Hosts a very large number of data sets
- Data is often available very rapidly

Earth Engine Data Catalog

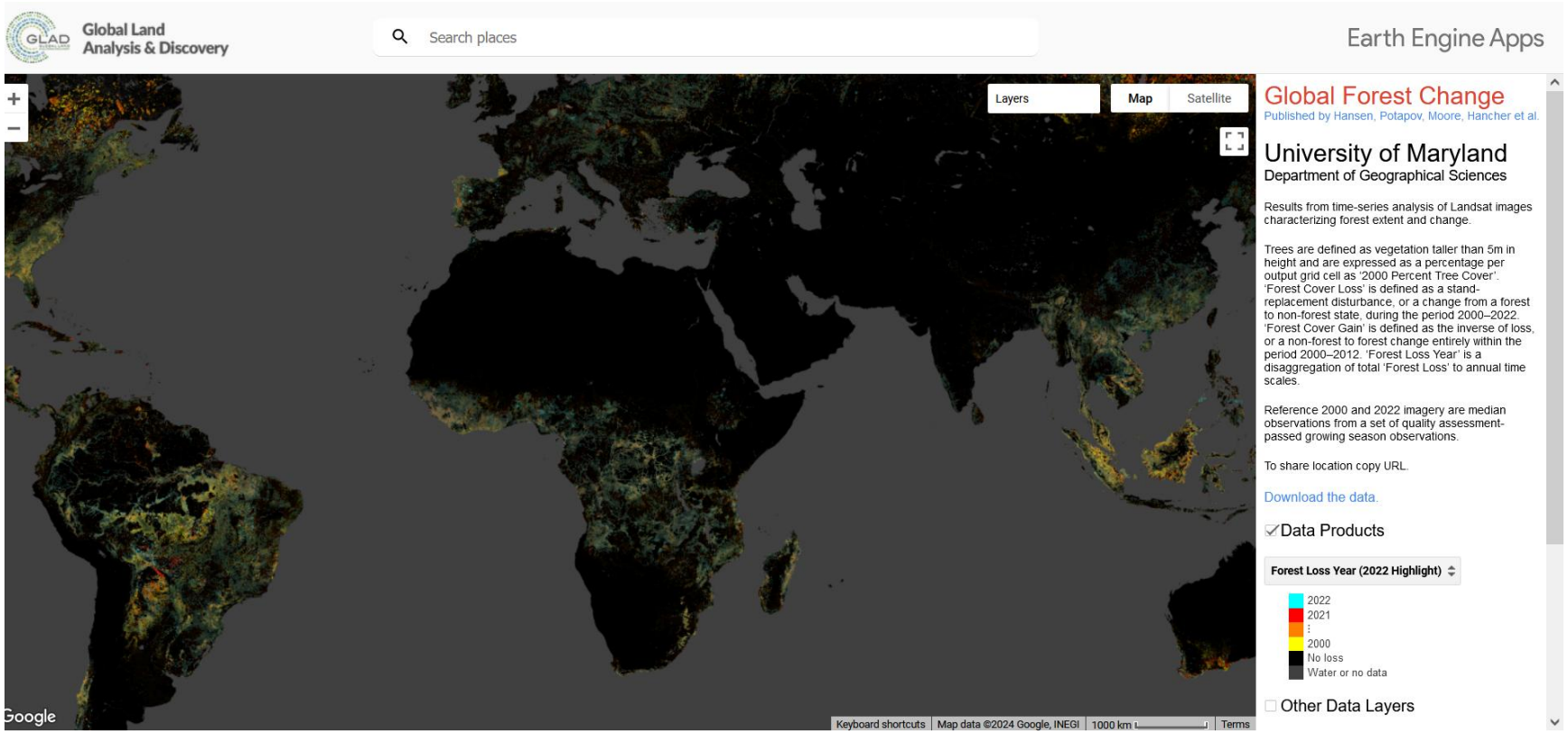


Google Earth Engine



Quick Demo: Global Forest Loss

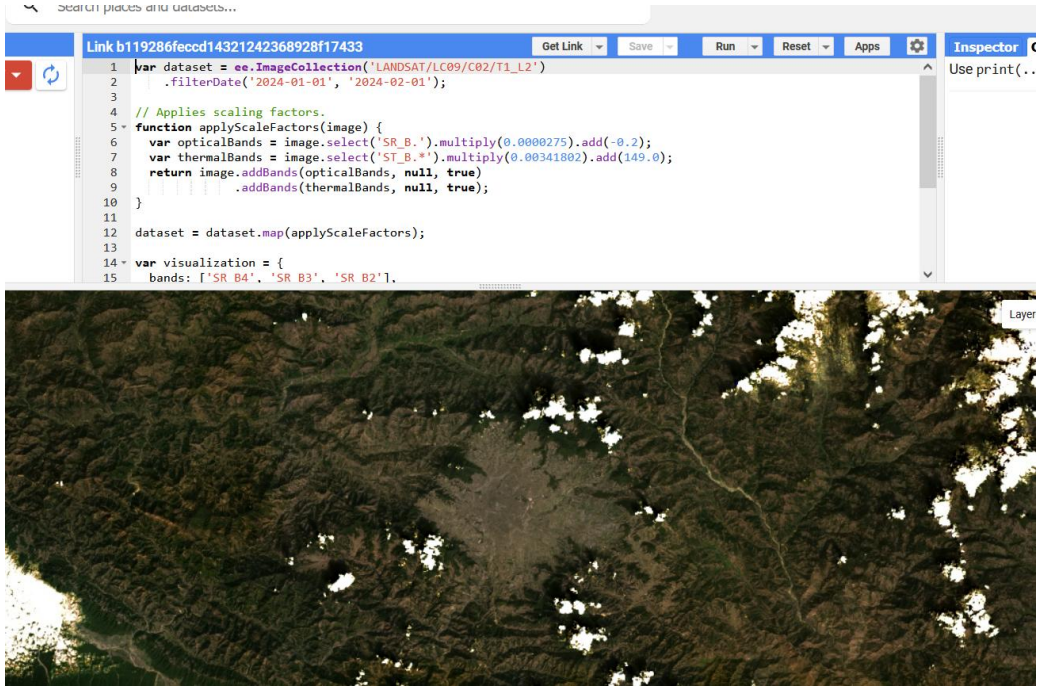
https://glad.earthengine.app/view/global-forest-change#bl=off;old=off;dl=1;lon=20;lat=10;zoom=3;



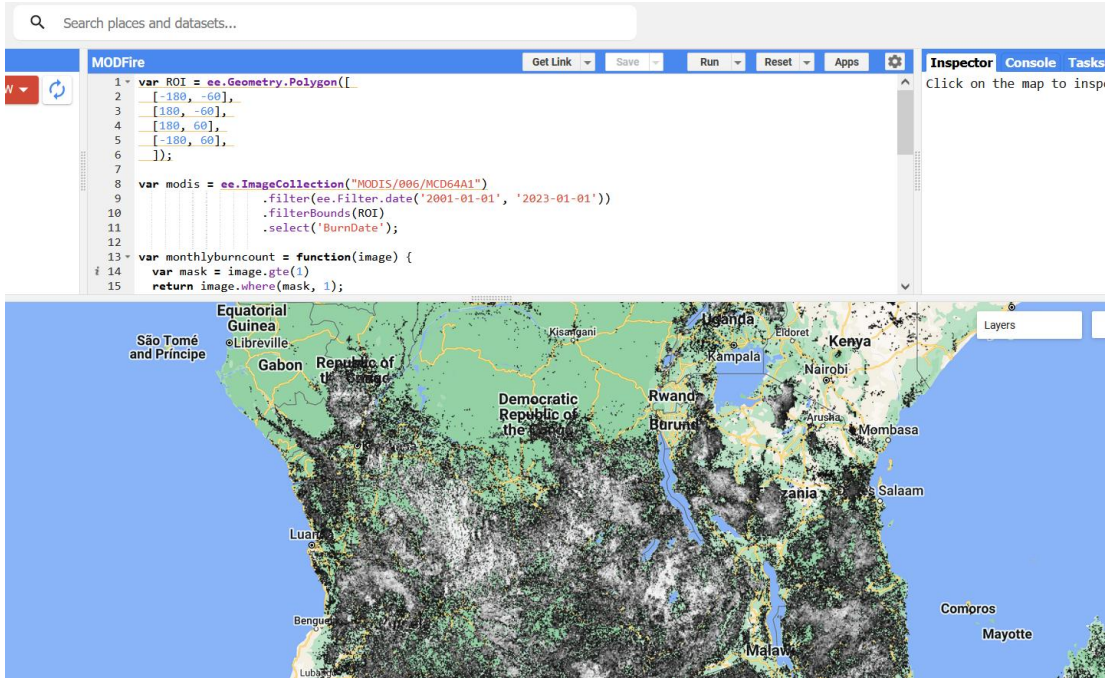
Quick Demo: LANDSAT Data

https://developers.google.com/earth-engine/datasets/catalog/LANDSAT_LC09_C02_T1_L2

https://code.earthengine.google.com/7434f942acf58ef9907d3b0b2216e304



Quick Demo: MODIS Fires

https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD64A1

https://code.earthengine.google.com/587a6b4b66e8733ffc0609841699e561